

## 2EXPERIMENT 2

### IDENTIFYING AN IDENTIFIER AND CLASSIFY IT AS VARIABLE, ARRAY, OR FUNCTION

**Aim:** To design and implement a transition diagram that recognizes identifiers and classifies them into

- Variable
- Array
- Function

**Design:**

**Algorithm:**

1. Read first character.
2. If it is letter or '\_'  
    Go to S1  
Else  
    Reject.
3. Read remaining letters/digits/\_.
4. Stay in S2.
5. When a delimiter is reached  
  
    if next symbol is '['  
        Array  
  
    else if next symbol is '('  
        Function  
  
    else  
        Variable

**Program:**

**Output:**

**Result:** The transition diagram for identifying identifiers was successfully designed and implemented using a finite automaton. The automaton correctly recognized valid identifiers and classified them as variables, arrays, or functions based on the subsequent symbol ([, (, or other delimiters). Invalid identifiers that did not satisfy the lexical rules were rejected successfully.